CommunicaTED: Communication on a Vertical Axis Using Cosmic Ray Muons

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Abstract

We propose a novel wireless telecommunication system that exploits the penetrating properties of atmospheric cosmic ray muons to transmit binary-encoded data through obstructions impenetrable to conventional electromagnetic signals — including solid rock and deep water. The system modulates the natural muon flux using a Helmholtz coil configuration to deflect charged muons in a controlled manner, encoding information as binary states detected by a downstream scintillator array. A Threshold Cherenkov veto system eliminates hadronic contamination to ensure readout fidelity. Analytical calculations predict a lateral muon deflection of approximately 1.106 cm over a 4-meter propagation distance for 4 GeV/c muons under a 0.1199 T field, with momentum spread effects modelled via Monte Carlo simulation. We further demonstrate thermal stability of the coil assembly, with temperature rise estimated at ~2°C/s under active operation, manageable through passive air cooling. Experimental validation is proposed at the CERN T9 beamline.

Keywords: *cosmic ray muons, muon communication, Helmholtz coil, binary encoding, particle detector, scintillator, Cherenkov veto*

1. Introduction

Communication infrastructure underpins modern civilisation, yet many critical scenarios remain beyond the reach of conventional electromagnetic (EM) technology. Radio and microwave signals are absorbed or reflected by dense geological formations, seawater, and reinforced structures, creating fundamental coverage gaps in deep-sea operations, underground mining, and through-mountain military communications.

Cosmic ray muons — highly penetrating charged leptons produced by the interaction of primary cosmic rays with the upper atmosphere — traverse kilometres of solid matter with negligible attenuation. This work proposes CommunicaTED: a binary wireless communication system that modulates the trajectory of atmospheric muons using a magnetic field to encode and transmit data through otherwise opaque media.

The operating principle is straightforward: a Helmholtz coil pair generates a controlled transverse magnetic field. When the field is active, passing muons are deflected onto one of two separated scintillator detectors (logical '1'); when inactive, muons propagate undeflected and miss both detectors (logical '0'). Varying the on/off duration of the coil field can further encode information in a Morse-code-like protocol.

Due to spatial constraints at the CERN test facility, experimental parameters differ from prospective real-world deployment values. A Python utility is provided (Section 3.3) to scale all calculations to arbitrary geometric and beam parameters.

2. Motivation and Scientific Case

2.1 Limitations of Conventional Communication

Electromagnetic signals in all practically used frequency bands — from low-frequency radio to millimetre-wave — experience exponential attenuation in conductive or high-density media. Seawater attenuates radio signals with a skin depth of only a few metres at ELF frequencies; rock provides similarly effective shielding. Acoustic and optical alternatives suffer comparable or worse limitations in such environments.

2.2 Why Muons?

Muons present a compelling alternative carrier medium for several reasons:

- Long mean lifetime (~2.2 μs in the rest frame), enabling relativistic travel over kilometre-scale distances before decay.
- High rest mass (~105.7 MeV/c^2; approximately 207 times the electron mass), yielding low radiative energy loss and enabling penetration of large material thicknesses with minimal deflection due to multiple Coulomb scattering.
- High natural flux at sea level (~1 muon cm^{-2} min^{-1}), ensuring a continuous, available signal carrier without artificial particle generation.
- Responsiveness to magnetic fields (charge ±e), enabling controllable beam deflection without requiring direct interaction with the transmission medium.

2.3 Why CERN?

The natural muon flux, while sufficient in principle, is irregular in energy and direction. Validation of the deflection mechanism requires a well-characterised, collimated muon beam with known momentum. The CERN T9 East Hall beamline provides muon beams with tunable momentum in the range of several GeV/c — precisely the regime relevant to this experiment. This controlled environment allows quantitative verification of deflection models before deployment with natural cosmic ray muons.

3. Experimental Methodology

3.1 Helmholtz Coil Design and Magnetic Field

A Helmholtz coil pair produces a highly uniform magnetic field in the central region between two identical circular coils of radius r separated by a distance equal to r. This geometry cancels the leading-order field gradient, providing a flat-field region well-suited to controlled particle deflection. External fields, including Earth's geomagnetic field (~50 μT), are simultaneously suppressed.

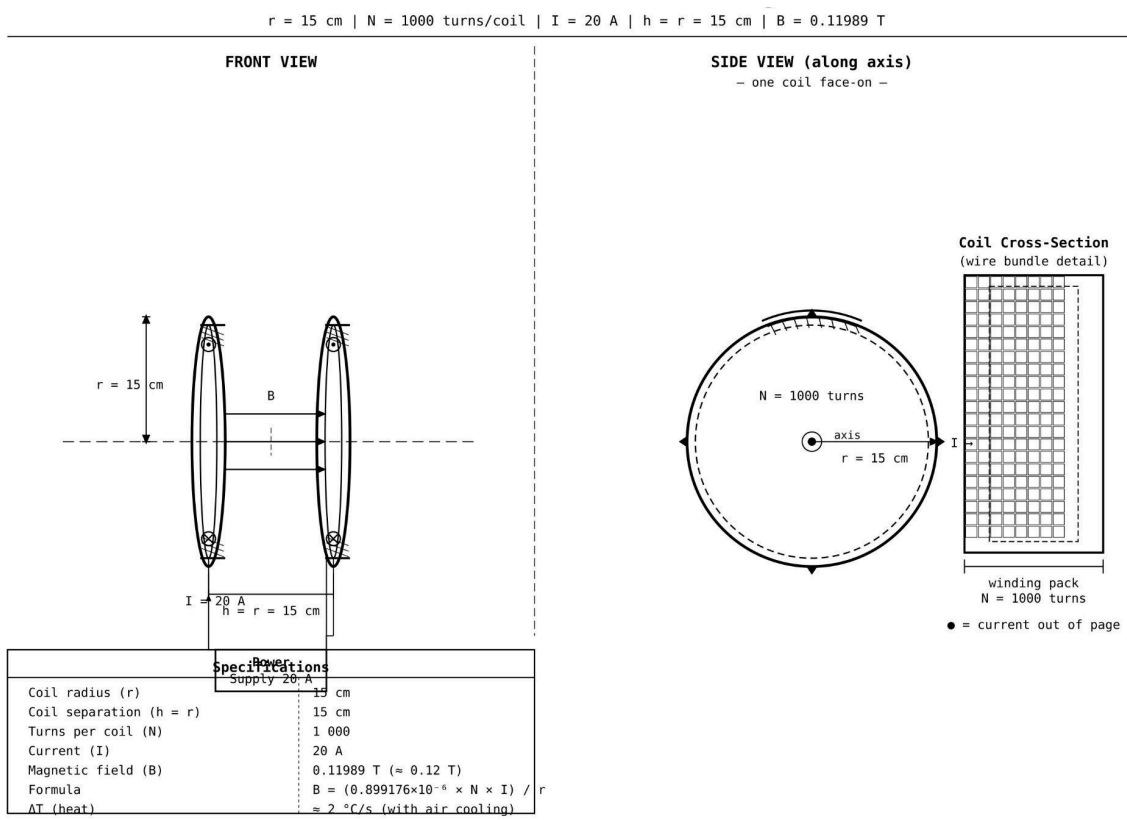
The central field magnitude for a Helmholtz configuration is given by:

$$B = (0.899176 \times 10^{-6} \times N \times I) / r$$

where N is the number of turns per coil, I is the applied current (A), and r is the coil radius (m). For our design parameters — r = 0.15 m, N = 1000, I = 20 A — the resulting field is:

$$B = (0.899176 \times 10^{-6} \times 1000 \times 20) / 0.15 = 0.11989\ T$$

This field strength, while modest compared to laboratory superconducting magnets, is sufficient to produce a detectable lateral deflection of GeV-scale muons over the available propagation distance, as shown in Section 3.2.



3.2 Thermal Analysis of Coil Assembly

Given that the coil assembly must be rapidly switched on and off to encode data, thermal stability under pulsed operation is a critical engineering constraint. The instantaneous temperature rise rate of a resistive coil carrying current I is derived by equating Joule heating to thermal energy stored in the wire volume:

$$\Delta T/\Delta t = (I^2 \rho_0) / (c \cdot d \cdot A^2)$$

where ρ_0 is the electrical resistivity of the conductor (copper: 1.68×10^{-8} Ω·m), c is the specific heat capacity (copper: 386 J/kg·K), d is the mass density (copper: 8960 kg/m³), and A is the wire cross-sectional area. For a wire gauge of $A = 645 \times 10^{-9}$ m² and I = 20 A:

$$\Delta T/\Delta t = (20^2 \times 1.68 \times 10^{-8}) / (386 \times 8960 \times (645 \times 10^{-9})^2) \approx 4.67\ °C/s$$

Incorporating surface heat dissipation reduces this effective rate to approximately 2°C/s. Given that individual encoding pulses are of short duration and passive air cooling is employed, thermal runaway is not a concern under normal operating conditions.

3.3 Muon Deflection Geometry

A muon of momentum p traversing a transverse magnetic field B follows a circular arc (cyclotron motion) with radius:

$$R = p / (q \cdot B)$$

where $q = 1.602 \times 10^{-19}$ C is the elementary charge. Using the conversion 1 GeV/c = 5.344286×10^{-19} kg·m/s, a 4 GeV/c muon has a momentum of $p = 21.377 \times 10^{-19}$ kg·m/s, yielding:

$$R = (21.377 \times 10^{-19}) / (1.602 \times 10^{-19} \times 0.11989) = 111.44 \text{ m}$$

Since R >> r (the coil radius of 0.15 m), the arc through the field region can be treated as a small perturbation. Using a similar-triangles geometric construction, the lateral displacement Δx accumulated over a 4-metre post-field propagation length is:

$$\sqrt{(R^2 - (2r)^2)} / d_{prop} = 2r / \Delta x_{inner}$$

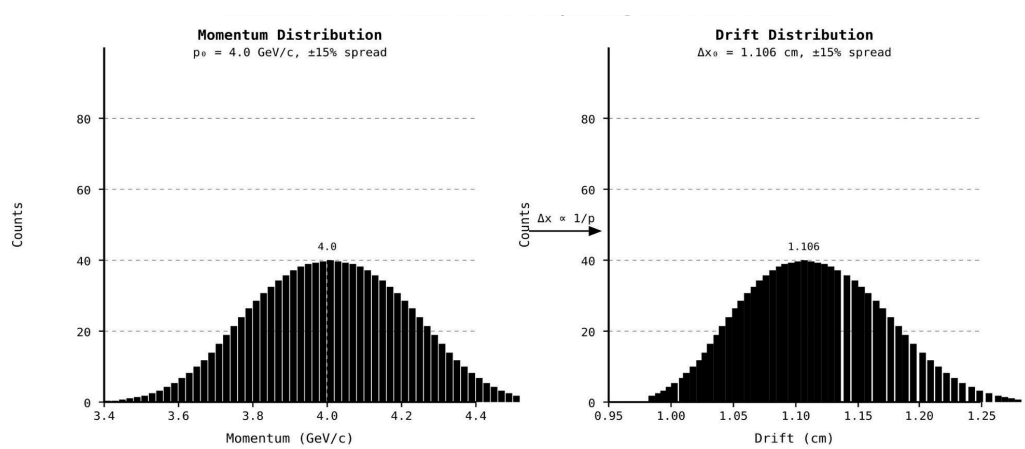
Numerically, this gives $\Delta x_{inner} = 1.076$ cm. Adding the additional geometric displacement R − A = 0.04 cm (where A is the chord length through the field region) yields a total transverse deflection of:

$$\Delta x_{total} = 1.076 + 0.040 = 1.116 \text{ cm over } 4 \text{ m}$$

This deflection is sufficient to direct muons from the beam axis onto separated scintillator panels, enabling the binary readout described in Section 3.4.

3.4 Momentum Spread and Statistical Modelling

The CERN beamline delivers muons with a nominal momentum of 4 GeV/c but with a ±15% RMS spread. Since the cyclotron radius R is inversely proportional to momentum, lower-momentum muons experience larger deflections and vice versa. The resulting distribution of lateral displacements, computed via Monte Carlo sampling of the beam momentum spectrum, is approximately Gaussian and centred near 1.10 cm, with standard deviation ~0.05 cm. All muons within the realistic momentum range are deflected beyond the 1 cm threshold required to reach the scintillator acceptance window, confirming robustness of the readout against beam momentum spread



3.5 Detection System and Binary Encoding

The detection system consists of two plastic scintillator panels positioned symmetrically about the beam axis at 4 m downstream of the coil assembly — one for positively charged muons (μ^+) and one for negatively charged muons (μ^-), which are deflected in opposite transverse directions by the applied field.

Binary encoding is implemented as follows: when the Helmholtz coils are energised, the deflected muon beam produces a coincident signal in one of the scintillators — interpreted as a logical '1'. When the coils are de-energised, muons propagate along the original beam axis between the detectors, producing no signal — logical '0'. A Morse-code-like protocol can be implemented by varying pulse duration to distinguish dashes and dots.

3.6 Hadronic Veto and Trigger Logic

The CERN secondary beam contains a non-negligible fraction of hadronic contaminants (primarily pions and kaons) that would produce false scintillator signals if not suppressed. Muon selection is achieved using a Threshold Cherenkov detector upstream of the coil assembly.

The Threshold Cherenkov is configured such that particles exceeding the medium's Cherenkov threshold — hadrons at the beam momentum, which have lower velocity than muons of equal momentum — emit a detectable Cherenkov signal. The trigger logic is configured to veto scintillator readout whenever a Threshold Cherenkov signal is present, ensuring that only muon events contribute to the data stream.

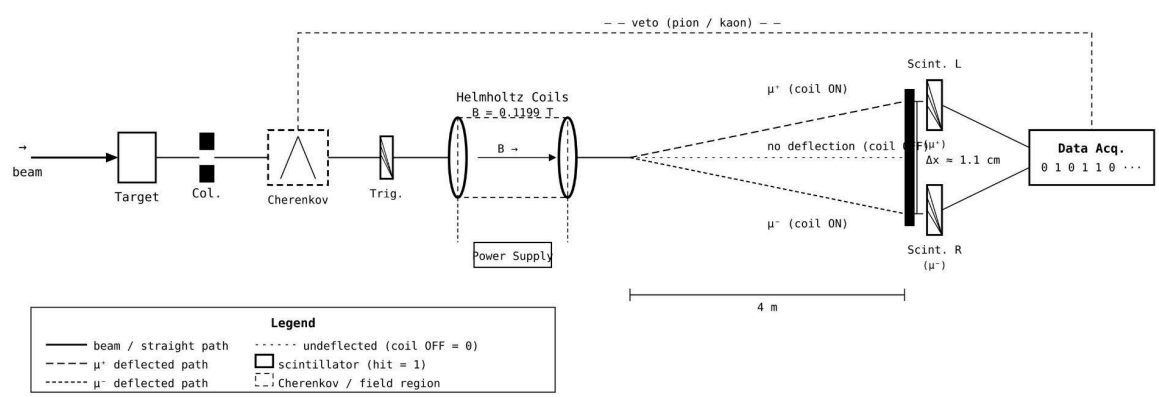
3.7 Python Calculation Utility

The following Python function computes the lateral deflection Δx (in cm) for arbitrary experimental parameters, enabling straightforward adaptation of these calculations to real-world deployment scenarios:

```
import math

def getDrift(d=4, I=20, r=0.15, N=1000, p=4):    k = 8.99176e-7    T = (I
* N * k) / r    R = (p * 5.344285) / (1.6 * T)    A = math.sqrt(R**2 - 4
* r**2)    X = d * 2 * r / A    return 100 * X  # Returns Δx in cm
```

4. Experimental Setup



The proposed beamline layout proceeds as follows, in order along the beam axis: (1) target and collimator to define the beam direction; (2) Threshold Cherenkov detector for hadronic veto; (3) trigger scintillator for timing; (4) Helmholtz coil pair (field length: 30 cm, radius: 15 cm) acting as the modulator; (5) electromagnetic shield; and (6) two offset scintillator panels (L and R) connected to a data acquisition system.

The total instrumented beam path from coil exit to detector plane is 4 metres, chosen to maximise the accumulated lateral displacement whilst remaining within the spatial constraints of the CERN East Hall test area. In a real-world deployment, this separation could be extended to many tens of metres, increasing the deflection (and thus signal contrast) proportionally.

5. Expected Outcomes and Impact

Successful execution of the proposed experiment would yield:

- A quantitative proof-of-concept for magnetically modulated muon-based binary communication, with measured deflection values compared against theoretical predictions.
- Characterisation of signal-to-noise performance under realistic beam conditions, including hadronic contamination and momentum spread.
- Assessment of achievable data rate, limited by the muon flux density and coil switching speed.
- Practical experience with particle detector technology, trigger logic design, and data acquisition — directly applicable to broader experimental physics careers.

In the longer term, this work could inform the development of communication systems for deep-sea scientific installations, underground mines, and geophysically obstructed environments where EM-based systems are impractical. The proposed Morse-code-like encoding protocol is a first step toward higher-bandwidth modulation schemes.

6. Acknowledgements

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